Sophia N. Wilson

PhD Fellow in Sustainable AI | SAINTS Lab, University of Copenhagen (UCPH)
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Research Profile

PhD fellow in sustainable artificial intelligence (AI) with two strands: technical, studying how AI systems use data, compute, and energy, and where it can be reduced; and socio-technical, measuring and mitigating the AI industry's sustainability harms, from environmental impact to poor labour conditions. Background in physics.

Education

PhD Fellow in Sustainable AI, SAINTS Lab, Dept. of Computer Science, UCPH. 2025 – present
Supervision: Raghav Selvan and Erik Dam. Funding: Independent Research Fund Denmark.

- **Costs of hyper-datafication:** led a study of the environmental, social, and economic costs of ever-larger datasets, combining a metadata analysis of 550,000 Hugging Face datasets, a survey of data-worker conditions in Kenya, and an analysis of data-centre investment and its geographic concentration (FAccT 2026).
- **Data frugality in practice:** led a study showing that claimed energy savings of data frugality are rarely measured, that data-frugal methods can cut both training energy consumption and bias, and proposed recommendations for how to close the preach–practice gap (ICML 2026, position paper track).
- **Algorithmic complexity of neural networks:** implemented experiments for QuBD (Quantized Block Decomposition), a scalable method for estimating the algorithmic complexity of network weights, giving the first scalable characterisation of the learning-as-compression hypothesis (NeurIPS 2026).

MSc in Computational Physics, Niels Bohr Institute, UCPH. 2023 – 2025
Thesis: "Quantifying the Reduction in Carbon Footprint of Physics-Informed Machine Learning". Supervision: Raghav Selvan and Jens Hesselbjerg Christensen. Resulted in a first-author paper on the trade-offs between performance and training and inference energy across architectural biases in shear flow forecasting (ECCV 2026, LIMIT Workshop).

BSc in Physics, Niels Bohr Institute, UCPH. 2019 – 2023
Thesis: "Measuring Cosmological Parameters with Fast Radio Bursts". Supervision: Kasper Heintz and Johan Fynbo.

Research Experience

Visiting Researcher, Samsung Semiconductor Denmark Research. Oct. 2026 – present
Research topic: profiling the hardware utilisation of agentic AI workloads on consumer hardware. Supervision: Vivek Shah. Funding: AI Denmark.

Research Assistant, Cosmic Dawn Center, Niels Bohr Institute, UCPH. Sep. 2022 – Jun. 2023
Research topics: gas, dust, and metal enrichment in high-redshift star-forming galaxies. Supervision: Kasper Heintz. Resulted in one first-author paper (A&A 2024) and three co-authored papers (A&A 2023, A&A 2023, ApJL 2023).

Awards

Best Poster (jury), Danish Digitalization, Data Science and AI 4.0 Conference. For "Characterizing Learning in Deep Neural Networks using Tractable Algorithmic Complexity Analysis" (NeurIPS 2026). Oct. 2026

Best Poster (public vote), Eastern European Machine Learning Summer School, Cetinje, Montenegro. For "Position: Stop Preaching and Start Practising Data Frugality for Responsible Development of AI" (ICML 2026). Aug. 2026

Grants & Funding

Samsung/UCPH Knowledge Collaboration, *co-applicant*, AI Denmark, 200,000 DKK. 2026

MSc thesis support, Stefan Rozental og Hanna Kobylinski Rozentals Fond, 20,000 DKK. 2024

Exchange semester at the University of Amsterdam, seven funders, 93,000 DKK. 2023 – 2024
Stefan Rozental og Hanna Kobylinski Rozentals Fond (30,000 DKK), Erasmus+ (20,500 DKK), Knud Højgaards Fond (12,000 DKK), Nordea-fonden (10,000 DKK), William Demant Fonden (10,000 DKK), Niels Bohr Fonden (5,500 DKK), Dansk Kvindesamfundslegater (5,000 DKK).

Teaching

Data Analysis and Machine Learning, *lecturer*, MSc course for high school teachers, Dept. of Computer Science, Aarhus University. 2026

Ethics and Technology, *guest lecturer*, BSc course, University College Copenhagen, "AI, Resource Use, and Sustainability". 2026

Machine Learning A, *teaching assistant*, MSc course, Dept. of Computer Science, UCPH. 2025

Machine Learning for Sciences, *guest lecturer*, PhD course, Dept. of Computer Science, UCPH, "Physics-Informed Machine Learning as a Carbon-Efficient Approach". 2025

Mathematics A, *teaching assistant*, high school, Falkonergårdens Gymnasium. 2025

Mathematics for Physicists 2, *teaching assistant*, BSc course, Niels Bohr Institute, UCPH. 2022 – 2023

Mathematics for Physicists 1, *teaching assistant*, BSc course, Niels Bohr Institute, UCPH. 2022 – 2023

Mechanics and Special Relativity, *teaching assistant*, BSc course, Niels Bohr Institute, UCPH. 2020 – 2021

Supervision

Two BSc students, *co-supervisor with Raghav Selvan*, Dept. of Communication, UCPH: "The Environmental Impact of Generative AI: A Life Cycle Assessment as a Basis for Studying Consumer Perception". 2026

Selected Talks and Panels

Brandenburgisches Zentrum für Medienwissenschaften, Potsdam, Germany, *invited talk*, "Hyper-Datafication and Its Environmental, Social, and Economic Costs". Oct. 2026

Halfspace, *invited talk*, "Assessing and Mitigating Sustainability Costs of Frontier AI". Oct. 2026

High Energy Physics Software Foundation Seminars, *invited talk*, "Assessing and Mitigating Sustainability Costs of Frontier AI". Sep. 2026

Center for AI, Danish School of Media and Journalism, *invited talk*, "Bæredygtig AI: Udfordringer og løsninger". Sep. 2026

Folkemødet, Bornholm, *panel discussion*, "Kærlighed, klimakrise og kunstig intelligens". Jun. 2026

Green AI Workshop, Pioneer Centre for AI, UCPH, *invited talk*, "How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI". May 2026

National Center for AI in Society, UCPH, *joint invited talk with Raghav Selvan*, "On the Challenges in Assessing the Sustainability of AI". May 2026

Lund Stem Cell Center, Lund University, *invited talk and panel discussion*, "How to Measure and Manage Energy Consumption of AI Models". Apr. 2026

Academic Service & Organisation

SAINTS Lab Journal Club, *coordinator*. 2026

D3A 4.0 Workshop on Resource-Aware and -Constrained AI, *co-organiser*. Oct. 2026

AAAI 2026 Fall Symposium on Parsimonious Scientific ML, *programme committee member*. Sep. 2026

BSc in Machine Learning, Dept. of Computer Science, UCPH, *admissions committee member*. Sep. 2026

SAINTS'26 Workshop on Sustainable AI for Sciences, *co-organiser and moderator*. Mar. 2026

Outreach

Working to inspire young people, especially girls and young women, to pursue STEM, currently through workshops and talks at high schools, e.g. on the carbon footprint of AI (Det Åbne Gymnasium, Sep. 2025) and on studying physics at university (Falkonergårdens Gymnasium, May 2025). Previously contributed to Women in Physics initiatives promoting gender balance in STEM (2021–2024) and volunteered at Insero's Summer Science Camp for primary school pupils (2023).

Media

Featured in Danish media on TV (TV2, Oct. 2026), on podcasts (Verden Forsøgt Forklaret, Sep. 2026; Indsigt, Nov. 2026), in newspapers (Uniavisen, Sep. 2026), and in online interviews (Women in Physics, Dec. 2025).

Publications

10 publications, 4 as first author. Titles link to the papers.

Peer-Reviewed Conference Proceedings

P. Bakhtiarifard, **S. N. Wilson**, M. Afifi, J. Wenshøj, and R. Selvan. Characterizing Learning in Deep Neural Networks using Tractable Algorithmic Complexity Analysis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.

S. N. Wilson, J. H. Christensen, and R. Selvan. Performance-Carbon Trade-Offs across Architectural Biases in Shear Flow Forecasting. In *Proceedings of the 19th European Conference on Computer Vision (ECCV), LIMIT Workshop*, 2026.

S. N. Wilson, A. Millard, G. F. Guðmundsdóttir, R. Selvan, and S. Mair. Position: Stop Preaching and Start Practising Data Frugality for Responsible Development of AI. In *Proceedings of the 43rd International Conference on Machine Learning (ICML), Position Paper Track*, 2026.

S. N. Wilson, S. Mair, M. Okinyi, E. B. Dam, J. Koch, and R. Selvan. How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026.

Peer-Reviewed Journal Articles

S. N. Wilson, K. E. Heintz, P. Jakobsson, S. C. Madden, D. Watson, G. Magdis, F. Valentino, T. R. Greve, and D. Vizgan. A High-Redshift Calibration of the [O I]-to-H I Conversion Factor in Star-Forming Galaxies. *Astronomy & Astrophysics*, 685, A30, 2024.

K. E. Heintz et al. (incl. **S. N. Wilson**, 11th of 41 authors). The Cosmic Buildup of Dust and Metals: Accurate Abundances from GRB-Selected Star-Forming Galaxies at $1.7 < z < 6.3$. *Astronomy & Astrophysics*, 679, A91, 2023.

K. E. Heintz et al. (incl. **S. N. Wilson**, 13th of 13 authors). Gauging the Mass of Metals in the Gas Phase of Galaxies from the Local Universe to the Epoch of Reionization. *Astronomy & Astrophysics*, 678, A30, 2023.

K. E. Heintz et al. (incl. **S. N. Wilson**, 30th of 30 authors). The Gas and Stellar Content of a Metal-Poor Galaxy at $z = 8.496$ as Revealed by JWST and ALMA. *The Astrophysical Journal Letters*, 944, L30, 2023.

K. E. Heintz et al. (incl. **S. N. Wilson**, 22nd of 23 authors). Spectroscopic Classification of a Complete Sample of Astrometrically-Selected Quasar Candidates Using Gaia DR2. *Astronomy & Astrophysics*, 644, A17, 2020.

Data Catalogues

K. E. Heintz et al. (incl. **S. N. Wilson**, 22nd of 23 authors). VizieR Online Data Catalog: Astrometrically-Selected QSO Candidates (Heintz+, 2020). *VizieR Online Data Catalog*, J/A+A/644/A17, 2021.